

Mohamed Tlili

GIS Consultant – Mineral Exploration | Geomatics Engineer

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Geomatics engineer specializing in GIS, cartography, and LiDAR processing, with expertise in geodatabase design, Python automation, and spatial data production for mineral exploration.

CORE COMPETENCIES

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| • GIS analysis and cartography | • Geodatabase design and data validation | • GIS-to-CAD conversion |
| • LiDAR point cloud classification | • Python automation and ArcGIS toolboxes | • Mineral tenure and claim analysis |
| • Photogrammetry (DTM/DSM, orthomosaics) | • Coordinate systems and datum transformations | • 3D modelling from point clouds |
| | | • Team leadership and project coordination |

PROFESSIONAL EXPERIENCE

Geomatics Specialist

EAGLEPROJECTS TUNISIA (subsidiary of EAGLEPROJECTS S.p.A., Italy) | Tunisia | Jul 2026 – Present

- Processed and registered point clouds from laser scanning and/or photogrammetric surveys.
- Cleaned, filtered, and optimized datasets.
- Aligned and georeferenced point clouds.
- Created 3D models, meshes, and surfaces from acquired data.
- Ensured structured management and storage of datasets.
- Collaborated with design teams (BIM, CAD, GIS).

Team Lead – Geodatabase Design and Integration

WOLO Engineering | Ariana, Tunisia | Dec 2025 – Present

- Led a team designing and deploying a cartographic geodatabase in ArcGIS Pro with custom data dictionaries; built automated data validation pipelines (Python and geodatabase rules) and integrated restitution deliverables for specification compliance.
- Coordinated project delivery and deadline adherence, ensuring team cohesion and continuous alignment with client requirements.

Cartographic Study Specialist

WOLO Engineering | Ariana, Tunisia | Dec 2023 – Dec 2025

- Processed and classified LiDAR point clouds (ground, vegetation, buildings) using MicroStation and TerraScan; generated digital terrain and surface models (DTM/DSM) and performed noise filtering.
- Conducted GCP placement and dense point cloud processing in Agisoft Metashape; generated DTM/DSM and produced high-resolution orthomosaics.
- Updated fiber-optic and infrastructure data in Smallworld GE; coordinated with cross-functional teams and clients to align spatial data with AutoCAD plans.
- Conducted precise digitization and feature extraction from drone imagery, ensuring the accuracy and reliability of cartographic deliverables.

FREELANCE EXPERIENCE

GIS Consultant (Freelance) – Mineral Exploration

Independent | Remote | Nov 2025 – Present

- Produced thematic maps, cartographic layouts, and technical documents for land analysis, mineral tenure, and geological exploration projects.
- Identified staking opportunities and detected claim overlaps by cross-referencing geological, cadastral, and tenure datasets (PLSS, BLM, Crown concessions, LTSA, mining claims) for targeted mineral exploration.
- Converted GIS data to CAD and created professional AutoCAD plans with complete cartographic styling, managing coordinate transformations and datum conversions in ArcGIS Pro.
- Automated repetitive geoprocessing and file management tasks using Python; built custom ArcGIS toolboxes to streamline spatial data workflows.

TECHNICAL SKILLS

GIS and Cartography: ArcGIS Pro, QGIS (advanced), Smallworld GIS, Agisoft Metashape

CAD: AutoCAD, MicroStation, FME

Programming: Python (geoprocessing, automation, toolboxes), LAS tools (LAStools), AutoLISP

PROJECTS

- **LAS File Manager:** Developed a Python application to split and merge LAS files by X/Y axes or grid, with interactive HTML map preview and real georeferencing; automated the workflow using LAStools for significant time savings in LiDAR data preparation.
- **Geodatabase Automation:** Automated batch import of 200+ CSV/Excel files into a geodatabase with field mapping and data normalization; built dynamic spatial attribute updates between shapefiles.
- **Geological Classification Tool:** Automated lithology and geological unit classification into 10 standard categories, ensuring data consistency and streamlined analysis.
- **UTM Grid Generator:** Generated UTM coordinate grids in AutoCAD for multiple coordinate systems and projections using a custom LISP script.

EDUCATION

National Engineering Diploma – Geomatics and Topography (EQF Level 7)

École Supérieure de l'Aéronautique et des Technologies (ESAT) | Ariana, Tunisia | 2020 – 2023

LANGUAGES

Arabic (Native) | French (Professional) | English (Professional)